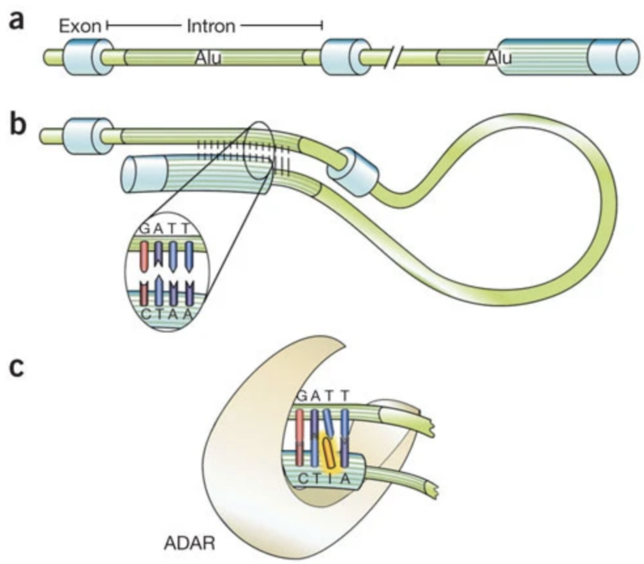
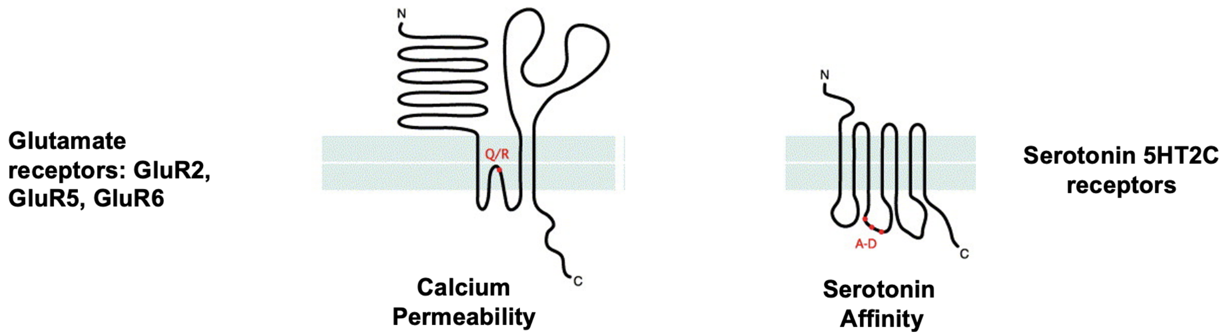
Neuro-genomics - class 7
Detecting DNA mutations and RNA editing events

The aim of this class, and the exercise that accompanies this class, is to demonstrate how modifications can be detected in RNA using RNAseq data. A very similar analysis can be performed to detect DNA modifications (for example somatic mutations) using DNAseq data, or even using RNAseq data. Here we will detect a specific type of RNA modifications - RNA editing events. We will use RNAseq data of the giant fiber lobe of the squid (the main ganglion of the squid) to detect A-to-I RNA editing events. Then, we will perform a similar analysis using RNAseq data of the gills of the squid. This will show that in the squid (and octopus as well), RNA editing is widespread in the nervous system, and not in other tissues, presumably in order to create protein complexity in the nervous system.

Reminder - RNA editing in human cells and tissues
RNA editing is the process in which the RNA is modified, via enzymes, in specific points compared to the information in the DNA. This can result in different protein sequences, thereby increasing the protein complexity beyond that of the genome. In human cells, the key mechanism for mRNA editing is via the enzyme ADAR, which binds double stranded RNA and changes Adenosine to Inosine. This is known as A-to-I editing. Image from this paper:



In the human brain, there are several examples of how RNA editing can increase the proteins complexity by creating proteins with amino acids that are not encoded in the genome:



Since Inosine is interpreted as Guanosine by the cell machinery and the sequencing machines, A-to-I editing is as if A is modified to G. Therefore, the computational detection of RNA editing is relatively simple - we need to align RNA reads against the genome, and check for locations in which A in the genome is modified into G in the RNA. Remember that this is possible only when full alignment of the reads is performed, using Bowtie2 or HiSAT2 for example, in contrast to pseudo-alignment such as Kallisto.

ACTAGTCAACCTCAGGGAATCA
ACTAGTCAACCTCAGGGAATCA
ACTAGTCAACCTCAGGGAATCA
ACTAGTCAACCTCAGGGAATCA
ACTAGTCAACCTCGGGGAATCA
ACTAGTCAACCTCGGGGAATCA
ACTAGTCAACCTCAGGGAATCA
ACTAGTCAACCTCGGGGAATCA
ACTAGTCAACCTCAGGGAATCA

Say that we are examining the aligned RNA reads in a given position, and suppose that we have 50 RNA reads aligned to that position overall. Say that the genome only shows A, and in the RNA we have 10 G and 40 A. Also suppose that the sequencing error rate is 0.1. The question now is what is the probability that this position shows evidence for RNA editing?

The binomial probability of that being a sequencing error is calculated by combining the probability of 10 "successes" (10 G) with the probability of 11, 12, 13 and so on until 50. binomial_CDF(9,50,0.1) is the probability of obtaining 9 or less "successes", so what we need is:

1-binomial_CDF(9,50,0.1)

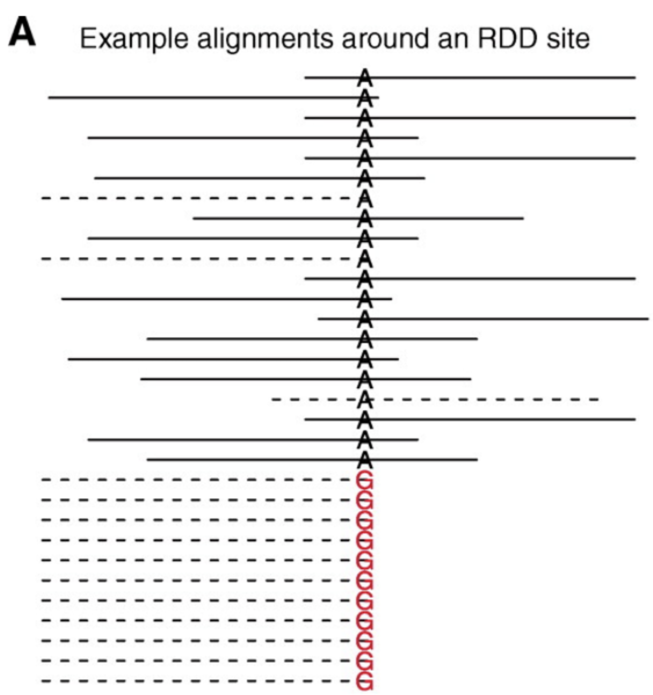
which is the same as binomial_CDF(40,50,0.9), but the latter is more accurate due to rounding issues.

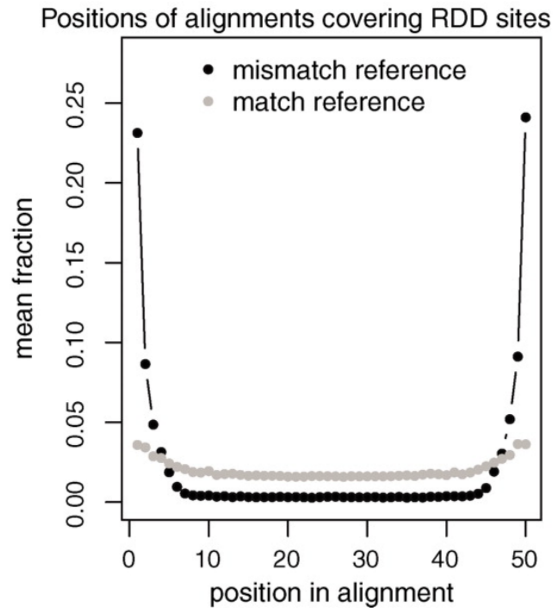
```
% Matalb code
>> binocdf(40,50,0.9)
ans =
   0.0245
```

Thus, the P-value is 0.0245, and using 0.05 level we can reject the null assumption (however, remember the multiple-testing problem).

Back in 2011, researchers started to use RNAseq to detect RNA modification in human cells. The first paper, which was published in Science, had the title “Widespread RNA and DNA Sequence Differences in the Human Transcriptome”. As the title suggests, the authors claimed that not only A-to-G modifications are observed in human cells, but rather all 12 possible modification types (A-to-C, C-to-A etc). It soon became clear that this conclusion resulted from artifacts in the analysis procedure.

As can be seen in the images from this paper (below), most of RNA and DNA Sequence Differences (RDD) were detected at the end of alignments between the RNA reads and the genome. This is a clear indication of alignment errors, which tend to happen at the end of the read:





Detecting RNA editing in squid ganglions
When we started the project of sequencing of the squid and the octopus, no genome was available for these species, and also the sequences of the genes was not known. However, the ease and relatively low cost of sequencing makes it so that characterization of a new species can be done quite easily. We generated RNA sequencing data of squid tissues, and the reads allowed us to construct the sequence of almost all the genes in the squid.

Step 1: De novo transcriptome assembly using Trinity
Input = paired-end reads of RNAseq, either 100 bases x2 or 150 bases x2
Output = the mRNA sequence of the genes
The general idea is illustrated in the figure below - i.e. the overlapping RNA reads can be used to generate gene sequence, even without alignment to the unknown genome.

```
AGTCTTCCTCGA
   CTTCCTCGAGATA
          GAGATACGATA
                  ATATACCGCC
                          CCATTTAGT
                               TAGTTTTTTGAG
                                       TGAGAGACG
                                             ACGCGCAGAGA
                                                    GAGAGGA

AGTCTTCCTCGAGATACGATATACCGCCCATTTAGTTTTTGAGAGACGCGCAGAGAGGA
```

mRNA sequence

However, the simple illustration is far from being realistic. The reason is that we expect extensive alternative splicing in the genes, and therefore several isoforms are expected for each gene. Trinity, and other de novo transcriptome assembly tools, are not trivial because they are making an attempt to detect the isoforms for each gene. When we detected the genes of the squid, we didn't care so much about the isoforms (because no genomics knowledge was available). Therefore, we just picked the longest 'isoform' for each gene and continued working with it.
The resulting genes were stored in a fasta text file termed '[pealeii.txt]'.

Step 2: Align the RNA reads and DNA reads against the genes
Because constructing a genome for species is difficult (mainly because of the size of the genome and the large fraction of repetitive elements in it), we couldn't generate a genome for the squid and the octopus. Therefore, the reads couldn't be aligned against the genome. Instead, we treated the genes as our reference and aligned the RNA and DNA reads against the genes. The steps to do that are listed below. Here we assume a Linux environment with bowtie2 installed.

First - use the fasta file with the gene sequences as if it contained the genome, and create indexes for it using bowtie2-build. The indexes will start with 'pealeii'
bowtie2-build pealeii.txt pealeii

Download the RNAseq of the squid giant fiber lobe
It can be downloaded from the NCBI SRA
Search for accession number: SRR1522987
In the previous classes, we explained how the fastq file of the sequencing reads can be downloaded from SRA.

Split the one paired-end fastq file into the two files, one for each paired-end read, as we demonstrated in the previous classes.
Below, the resulting two files are termed 'Squid_Pair1.fastq' and 'Squid_Pair2.fastq'

Then - run bowtie2 with local alignment and align the paired-end reads against the genes
The command should look like this: (note that here 12 threads are utilized, change this number according to the number of threads in your computer)
bowtie2 -x pealeii --local --threads 12 -1 Squid_Pair1.fastq -2 Squid_Pair2.fastq -S aligned_RNAreads.sam

The output of bowtie2 is a SAM file with all the alignments, and also summary statistics which is displayed on the screen:
21561343 reads; of these:
  21561343 (100.00%) were paired; of these:
    7579886 (35.15%) aligned concordantly 0 times
    13268351 (61.54%) aligned concordantly exactly 1 time
    713106 (3.31%) aligned concordantly >1 times

7579886 pairs aligned concordantly 0 times; of these:
  412952 (5.45%) aligned discordantly 1 time

7166934 pairs aligned 0 times concordantly or discordantly; of these:
  14333868 mates make up the pairs; of these:
    13478264 (94.03%) aligned 0 times
    435955 (3.04%) aligned exactly 1 time
    419649 (2.93%) aligned >1 times
68.74% overall alignment rate

Step 3: Use samtools to process the resulting SAM file

Reminder SAM/BAM format:

A compact way to pack all the information about the alignment into a text file.

Say that we have these following alignments:

```
Coor     12345678901234  5678901234567890123456789012345
ref      AGCATGTTAGATAA**GATAGCTGTGCTAGTAGGCAGTCAGCGCCAT

+r001/1        TTAGATAAAGGATA*CTG
+r002         aaaAGATAA*GGATA
+r003       gcctaAGCTAA
+r004                     ATAGCT..............TCAGC
-r003                            ttagctTAGGC
-r001/2                                        CAGCGGCAT
```

In SAM format then converts the alignment into this:

```
r001   99 ref  7 30 8M2I4M1D3M = 37  39 TTAGATAAAGGATACTG *
r002    0 ref  9 30 3S6M1P1I4M *  0   0 AAAAGATAAGGATA    *
r003    0 ref  9 30 5S6M       *  0   0 GCCTAAGCTAA       * SA:Z:ref,29,-,6H5M,17,0;
r004    0 ref 16 30 6M14N5M    *  0   0 ATAGCTTCAGC       *
r003 2064 ref 29 17 6H5M       *  0   0 TAGGC             * SA:Z:ref,9,+,5S6M,30,1;
r001  147 ref 37 30 9M         =  7 -39 CAGCGGCAT         * NM:i:1
```

The SAM format includes all the original information, including the gaps in the alignments, and the locations of mismatches.

Working directly with SAM files is not trivial. Therefore we will use two packages: the first is called samtools.

Install samtools

Then run these following commands, which will convert the SAM file into a more compact binary file, sort the binary file and create an index for it, respectively:

samtools view -S -b aligned_RNAreads.sam > aligned_RNAreads.bam

samtools sort aligned_RNAreads.bam -o aligned_RNAreads_sorted.bam
samtools index aligned_RNAreads_sorted.bam

The two resulting files can be found here:
aligned_RNAreads_sorted.bam
aligned_RNAreads_sorted.bam.bai

The second package that we will need to analyze the SAM files using python is called pysam, and it is basically a version of samtools designed for easy integration within python.
#Install pysam

Step 4: Repeat steps 1-3 for the squid gills (which we will compare to the squid giant fiber lobe)
Download the RNAseq of the squid gills
It can be downloaded from the NCBI SRA
Search for accession number: SRX817963
In the previous classes, we explained how the fastq file of the sequencing reads can be downloaded from SRA.

Split the one paired-end fastq file into the two files, one for each paired-end read, as we demonstrated in the previous classes.
Below, the resulting two files are termed 'Squid_gills_Pair1.fastq' and 'Squid_gills_Pair2.fastq'

Then - run bowtie2 with local alignment and align the paired-end reads against the genes
The command should look like this: (note that here 12 threads are utilized, change this number according to the number of threads in your computer)
bowtie2 -x pealeii --local --threads 12 -1 Squid_gills_Pair1.fastq -2 Squid_gills_Pair2.fastq -S aligned_RNAreads_gills.sam

```
22859774 reads; of these:
  22859774 (100.00%) were paired; of these:
    14153621 (61.91%) aligned concordantly 0 times
    8356904 (36.56%) aligned concordantly exactly 1 time
    349249 (1.53%) aligned concordantly >1 times
    ----
    14153621 pairs aligned concordantly 0 times; of these:
      1389042 (9.81%) aligned discordantly 1 time
    ----
    12764579 pairs aligned 0 times concordantly or discordantly; of these:
      25529158 mates make up the pairs; of these:
        22947840 (89.89%) aligned 0 times
        1715071 (6.72%) aligned exactly 1 time
        866247 (3.39%) aligned >1 times
49.81% overall alignment rate
```

Then run these following commands, which will convert the SAM file into a more compact binary file, sort the binary file and create an index for it, respectively:
samtools view -S -b aligned_RNAreads_gills.sam > aligned_RNAreads_gills.bam
samtools sort aligned_RNAreads_gills.bam -o aligned_RNAreads_gills_sorted.bam
samtools index aligned_RNAreads_gills_sorted.bam

The two resulting files can be found here:
aligned_RNAreads_gills_sorted.bam
aligned_RNAreads_gills_sorted.bam.bai

Step 5: Run jupyter scripts to analyze the data, and to detect RNA editing events
Here we provide two scripts for partial analysis of the giant fiber lobe (the nervous system) and the gills. In the exercise you will write scripts for a more complete analysis:
Analysis of one gene - Spectrin, a gene important for the cytoskeleton, and is linked for the formation of synapses in the brain
Class_editing_analysis.ipynb (the giant fiber lobe)
Class_editing_analysis_gills.ipynb (the gills)

Below, we will list several points regarding these jupyter scripts:

1) The concept of unique alignment. Recall that for each aligned read there is a score, termed mapping quality, which tells if the read is likely to align to a different location in the reference. The maximal mapping quality means unique alignment, i.e. the read doesn't not align to any other location of the reference. With bowtie2 this number is 42. So if we want to work only with reads with unique alignment, we will demand a mapping quality of 42.
2) In a fastq format, for each base there is a quality score, which is connected to the likelihood of a sequencing error. The higher the quality score, the less likely is the chance of a sequencing error. Quality score 30 corresponds to 1/1000 error rate, or 0.1% of sequencing error

$$Q = -10\ \log_{10} P$$

or

$$P = 10^{\frac{-Q}{10}}$$

3) NumPy array vs list in python. Numpy arrays are somehow similar to vectors in Matlab/R. This means that one can, for example, multiply all the values in the array by a factor, divide them, sum them etc. This is useful as it can not be done with standard lists in python.
4) The concept of pileup with samtools/pysam:

Pileup means that for each region in the reference (chromosome or gene in our case), we pile-up all the reads that match to that region and then go column-by-column (i.e. every base in the reference) and examine which reads are aligned to it, and if these reads show modifications (red bases above) compared to the reference. The pileup method of examining SAM files is useful for detecting DNA/RNA modifications.